

Fátima Miranda Pestaña



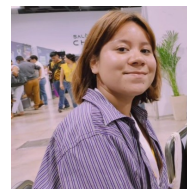
Data Scientist and Engineer

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Data professional with a strong foundation in data analysis, data engineering, data science, machine learning, natural language processing (NLP), and full-stack web development. Currently pursuing graduate studies in geospatial information science, with applied research experience at CentroGeo. Experienced in designing reproducible data workflows, building ML pipelines, and deploying scalable solutions on AWS and Google Cloud Platform. Proficient in Python, SQL/NoSQL database management, and C1-level English. Passionate about using data for social good, with active participation in international challenges, scientific conferences, and applied research.

Data Analytics

Data cleaning and preprocessing, exploratory and statistical analysis, dashboards and reporting. Python (Pandas, NumPy), SQL, Power BI (3+ years), Tableau, Looker Studio, Matplotlib, Seaborn.

Data Engineering

ETL pipeline design and automation; management and administration of relational and NoSQL databases (PostgreSQL, MySQL, SQLite, Snowflake, Spark); data standardization (XML, JSON, GeoJSON) and workflow orchestration. Dataiku, REST APIs, Git, CI/CD basics.

Data Science, Machine Learning & Deep Learning

Machine learning model design and evaluation (classification, KNN), cross-validation, feature engineering, and applied research. Python, Scikit-learn, Pytorch, Tensorflow.

Web Development (Front-end & Back-end)

Interactive applications and dashboards, UI/UX for non-technical audiences. React, Node.js, Express.js, Streamlit, JavaScript, HTML, CSS.

Cloud Computing

Cloud architecture, deployment, and automation. AWS (S3, EC2, Lambda, SageMaker — AWS Academy Graduate), GCP (BigQuery, Cloud Storage, Cloud Console), Azure (Data Factory, Blob Storage).

GIS — Geospatial Information Systems

Geospatial data analysis and visualization, remote sensing, and interactive mapping, with applied use of QGIS and Google Earth Engine. Backed by geospatial research experience at CentroGeo.

EDUCATION

Master's in Geospatial Information Sciences (in progress) 2025 – Present

CentroGeo — Centro de Investigación en Ciencias de Información Geoespacial, A.C. Mérida, Yucatán, México.

Graduate studies in applied geospatial analysis and geographic information systems. Thesis work focused on natural language processing (NLP) for low-resource languages, applied to Yucatec Maya.

Member of the GILTIA research group (Research Group on Languages, Territory, and Artificial Intelligence, CentroGeo). [View GILTIA](#)

B.S. in Data Engineering

Sep 2021 – Jul 2025

Universidad Politécnica de Yucatán. Mérida, Yucatán, México.

Coursework in data processing, machine learning, NLP, databases, and geospatial analysis. Key projects: an AI-powered deforestation monitoring system using drone imagery and KNN, social network analysis to detect influence patterns, and a classifier for indigenous language texts.

PUBLICATIONS

Molina-Villegas, A., Fernández-Sabido, S., Mendoza-Vargas, E., & Miranda-Pestaña, F. (2025). Detecting indigenous languages: A system for Maya text profiling and machine learning classification techniques. *International Journal of Computer and Information Engineering*, 19(1). [View publication](#)

EXPERIENCE

Data Scientist / CentroGeo, A.C. Ago 2024 – Dic 2024 · Mérida, Yucatán

Led development of a machine learning classifier for automatic language identification (Spanish vs. Yucatec Maya), validated through cross-validation and confusion matrices.

Engineered linguistic features that improved model performance in low-resource settings.

Deployed an interactive web demo for public testing and engagement with language technology.

Data Engineer Intern / CentroGeo, A.C. Feb 2024 – Jul 2024 · Mérida, Yucatán

Designed data-cleaning pipelines for text, tabular, and XML formats; parsed and transformed XML with Python to unify metadata across research projects.

Integrated the OpenAI API to automate Spanish–English translation with Mayan-language pivots, cutting manual translation effort by 60%.

Improved dataset processing speed and format compatibility by 30% through optimized routines.

Data Engineer / PLENUMSOFT May 2024 – Ago 2024 · Mérida, Yucatán

Developed Python scripts to automate the collection and processing of risk-related data across the US, Canada, and Mexico, in collaboration with academic partners.

Used Dataiku to automate and scale data management workflows, reducing data retrieval time by 40%.

Data Science Intern / EMBISOFT Engineering Mar 2023 – Ago 2023 · Mérida, Yucatán

Used Google Cloud Console to manage and cleanse large-scale datasets, improving accuracy and accessibility for business analysis.

LANGUAGES

Spanish: Native

English:

ITEP (International Test of English Proficiency)

Level C1 — Advanced

Nov 2024

[View credential](#)

TOEFL ITP

Score 560 — Level B2 (Intermediate)

May 2025

[View credential](#)

CERTIFICATIONS

1. AWS Academy Graduate — AWS Academy Cloud Foundations

Skills: AWS Architecture, Cloud, Core Services, Pricing, Support

Nov 2024

[View credential](#)

2. Google Project Management Certificate

Coursera (Google Career Certificates)

Aug 2024

[View credential](#)

3. Dataquest — Data Science Learning Platform

Completed courses: Data Modeling, API Management, Python Programming

2023–2024

4. U.S. Embassy in Mexico — Programa de Capacitación Especializada

Ethics and Environmental Awareness for the ICT Sector

Jun 2023

5. Gestión de Proyectos con Metodologías Ágiles y Enfoques Lean

Fundación Telefónica Movistar México

Jan 2024

+ Additional certifications available on my LinkedIn profile: [View profile](#)

Contact me:

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